EXPERIMENT – 5

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- *The user fills out a web form with their name and email.*
- *The form uses both GET and POST methods for data submission.*

2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. Create an HTML Form:

- *Design an HTML form with two input fields:*
  - *Name*
  - *Email*
- *Create two buttons to submit the form:*
  - *One using GET*
  - *One using POST*
- *Use the action attribute to define the Servlet URL and specify the form method.*

2. *Create the Servlet to Handle GET and POST Requests:*
   - *Write a Java Servlet class to handle both GET and POST requests.*
   - *The servlet will contain:*
     - *doGet() method to handle GET requests*
     - *doPost() method to handle POST requests*
   - *In both methods, retrieve form data using request.getParameter().*

3. *Handle GET Requests in the Servlet:*
   - *In doGet(), retrieve:*
     - *Name*
     - *Email*
   - *Display the data in the response.*
   - *The data will be visible in the URL.*

4. *Handle POST Requests in the Servlet:*
   - *In doPost(), retrieve:*
     - *Name*
     - *Email*
   - *Display the data in the response.*
   - *The data will not be visible in the URL (sent in request body).*

5. *Return the Output:*
   - *The Servlet returns an HTML page displaying:*

    - *Name*
    - *Email*
- *For GET → data appears in URL*
- *For POST → data appears only in response body*

6. Deploy and Test the Servlet:

- *Compile the Servlet class.*
- *Deploy it on a Servlet container (e.g., Apache Tomcat).*
- *Run the application.*

7. Observe and Compare Results:

- *GET Method:*
    - *Data is sent through the URL*
- *POST Method:*
    - *Data is sent through the HTTP request body*

CODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Bus Pass - GET vs POST</title>

<style>
body{
   margin:0;
   font-family: Arial;
```

```
    background: linear-gradient(135deg,#1e3c72,#2a5298);
    color:white;
}

header{
    background:#1e88e5;
    padding:15px;
    text-align:center;
    font-size:24px;
    font-weight:bold;
}

nav{
    background:#0d47a1;
    padding:10px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
    margin:0 15px;
    font-weight:bold;
}

.container{
    width:420px;
```

```
    margin:50px auto;
    padding:25px;
    background: rgba(255,255,255,0.15);
    border-radius:12px;
    box-shadow:0 0 15px rgba(0,0,0,0.3);
}

h2{
    text-align:center;
}

.form-group{
    margin-bottom:15px;
}

input{
    width:100%;
    padding:8px;
    border:none;
    border-radius:5px;
}

button{
    width:100%;
    padding:10px;
    border:none;
    border-radius:25px;
```

```
    background: linear-gradient(45deg,#00e5ff,#00c853);
    color:white;
    cursor:pointer;
    margin-top:10px;
}

.section{
    margin-top:20px;
}
</style>

</head>

<body>

<header>
    Online Bus Pass Application System
</header>

<nav>
    <a href="#">Home</a>
    <a href="#">Bus Routes</a>
    <a href="#">Bus Stops</a>
</nav>

<div class="container">
```

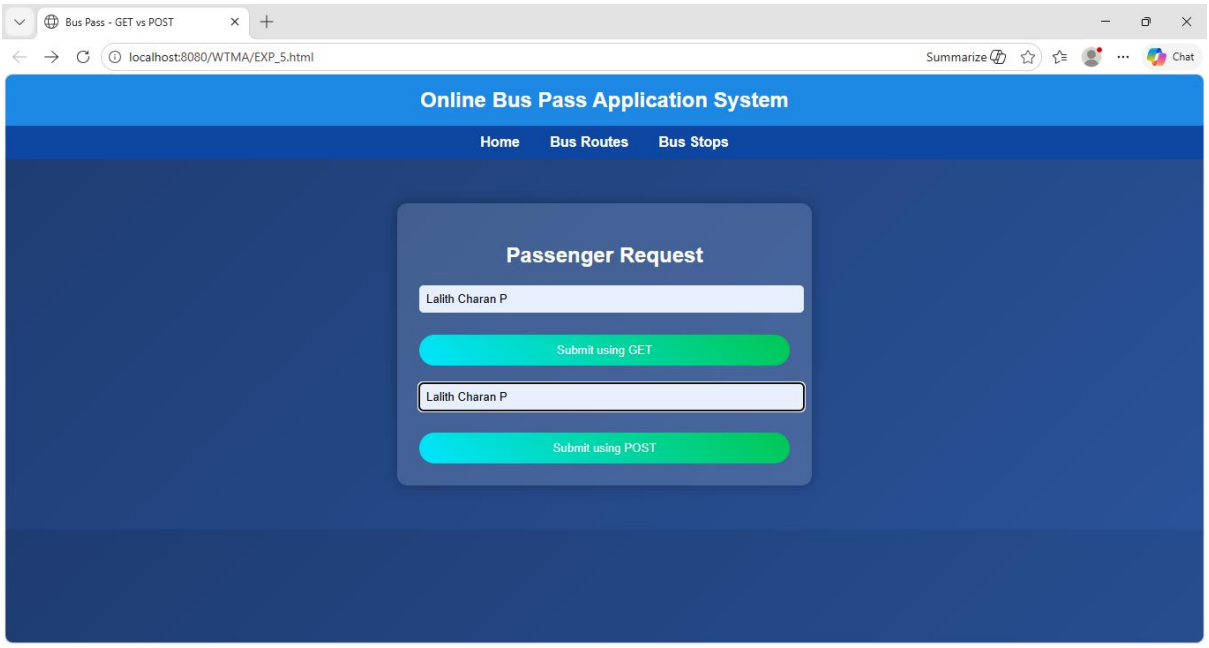
```
    <h2>Passenger Request</h2>

    <!-- GET FORM -->
    <div class="section">
        <form action="EXP5Servlet" method="get">
            <div class="form-group">
                <input type="text" name="username" placeholder="Enter
Name" required>
            </div>
            <button type="submit">Submit using GET</button>
        </form>
    </div>

    <!-- POST FORM -->
    <div class="section">
        <form action="EXP5Servlet" method="post">
            <div class="form-group">
                <input type="text" name="username" placeholder="Enter
Name" required>
            </div>
            <button type="submit">Submit using POST</button>
        </form>
    </div>

</div>

</body>
</html>
```

```
package WTMA;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;
import java.io.*;

@WebServlet("/EXP5Servlet")   // ✓EXACT match with HTML
public class EXP5SERVLET extends HttpServlet {

    private void generateResponse(PrintWriter pw, String name, String
method) {

        pw.println("<html><head><title>Result</title>");

        pw.println("<style>");
        pw.println("body{margin:0;font-family:Arial;background:linear-
gradient(135deg,#1e3c72,#2a5298);color:white;text-align:center;}");
        pw.println("header{background:#1e88e5;padding:15px;font-
size:22px;}");
        pw.println(".card{margin:100px
auto;width:400px;padding:30px;border-
radius:12px;background:rgba(255,255,255,0.15);box-shadow:0 0 15px
rgba(0,0,0,0.3);}");
        pw.println(".icon{font-size:35px;margin-bottom:10px;font-
weight:bold;}");
        pw.println("</style>");

        pw.println("</head><body>");
```

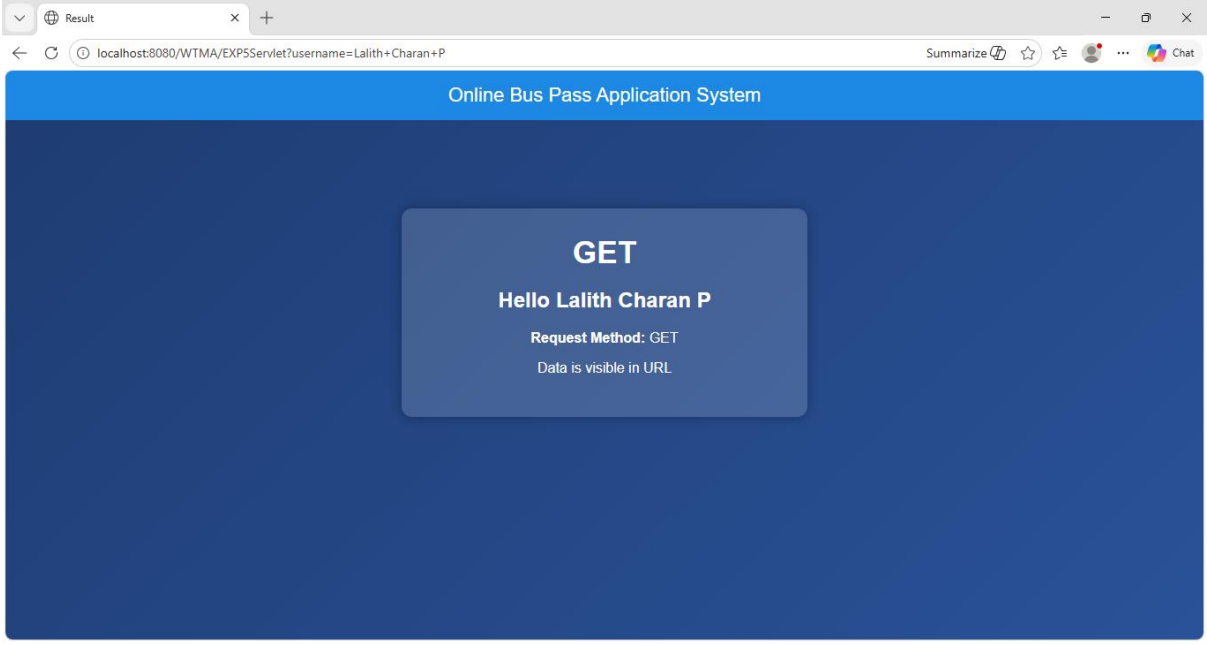
```
        pw.println("<header>Online Bus Pass Application
System</header>");

        pw.println("<div class='card'>");
        pw.println("<div class='icon'>" + method + "</div>");
        pw.println("<h2>Hello " + name + "</h2>");
        pw.println("<p><b>Request Method:</b> " + method + "</p>");

        if(method.equals("GET")){
            pw.println("<p>Data is visible in URL</p>");
        } else {
            pw.println("<p>Data is hidden from URL</p>");
        }

        pw.println("</div>");
        pw.println("</body></html>");
    }

    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter pw = response.getWriter();

        String name = request.getParameter("username");
```

```
        generateResponse(pw, name, "GET");
    }

    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter pw = response.getWriter();

        String name = request.getParameter("username");

        generateResponse(pw, name, "POST");
    }
}
```

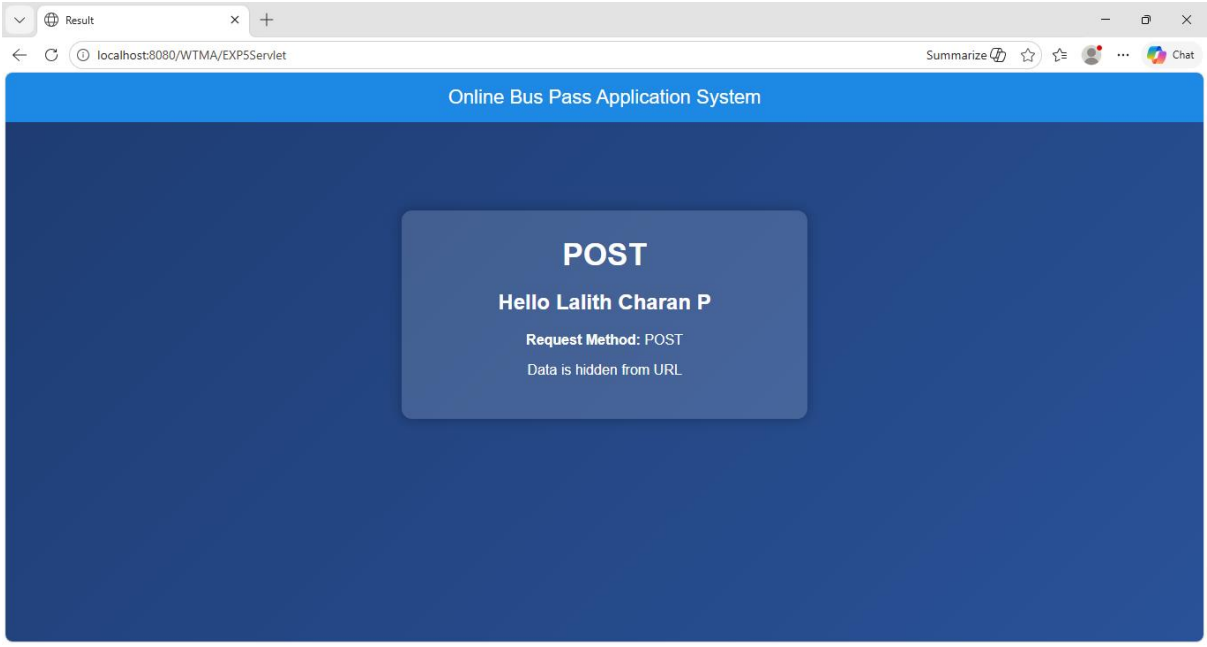
OUTPUT 1:



OUTPUT 2:



OUTPUT 3:



RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.